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Espacio ℓ^p

Definición 1 (Espacio ℓ^p). Sea $(\mathbb{N}, \mathcal{P}(\mathbb{N}), |\cdot|)$ el espacio de conteo y $p \in [1, \infty]$,

$$\ell^p := \mathcal{L}^p(\mathbb{N}, \mathcal{P}(\mathbb{N}), |\cdot|) = \left\{ \overline{a} = (a_n)_{n \in \mathbb{N}} \in \mathbb{K}^{\mathbb{N}} : \|\overline{a}\|_p < \infty \right\}$$

$$\text{donde } \forall \overline{a} \in \mathbb{K}^{\mathbb{N}} : \|\overline{a}\|_p = \begin{cases} \left(\sum_n |a_n|^p\right)^{1/p}, & p < \infty, \\ \sup_n |a_n|, & p = \infty. \end{cases}$$